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Research papers

Simultaneous internal heating for balanced temperature and state-of-charge distribution in lithium-ion battery packs

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ABSTRACT

Lithium-ion battery cells are widely known to work reliably in a certain range of temperatures. While operating in a high-temperature condition might lead to thermal runaway, they also suffer significant performance degradation and power/energy capacity loss in sub-zero temperatures. Consequently, heating the cells before usage plays a critical role in battery-powered applications. Among different heating methods, one promising approach is internal mutual pulse heating which offers many advantages as it utilizes the internal resistance of the battery cell to create heat and increase the cell temperature. However, in the context of a large battery pack, the nonuniform distribution of temperature and State-of-Charge (SOC) between the cells heavily affect the performance of the battery pack and remain to be unsolved. In this study, we present a internal heating technique for simultaneously balancing temperature and SOC of the battery cells considering the geometrical and thermal characteristics of the battery pack without requiring the characterization of the cell impedance. The proposed technique schedules the order and timing of mutual pulse heating for different groups of cells in a battery pack to reduce the variance in temperature and SOC during and after pre-heating. The simulation results showed that the SOC and temperatures among the cells can be kept balanced while the average pack temperature is maintained around a target temperature during a long period in a wide range of temperatures below zero degrees Celsius.

1. Introduction

Lithium-ion batteries have been widely adopted as the main power source in many consumer products as well as industrial applications. Thanks to many advantages such as high energy density and having virtually no memory effects, they enable the development of portable devices for operating in remote or hazardous areas where grid power is not available. The advantages of lithium-ion batteries are obvious but also are quickly diminished when being used in undesirable conditions.

In sub-zero temperatures, lithium-ion batteries suffer significant degradation in terms of performance and lifespan [1]. For instance, when the cell temperature is −10 °C, the discharge capacity of a 2.2 Ah cylindrical cell reduced to 1.7 Ah at 1*C* discharge rate and only about 0.9 Ah at 4.6*C* discharge rate. [2]. At −20 °C, it was shown that a lithium $LiFePO_4M_n$ battery can only deliver 60.6% of its nominal power [3]. Meanwhile, a phenomenon called lithium plating is likely to incur in low temperature where metallic lithium forms at the anode-separator interface of the cell during charging, resulting in malfunction over time and a shorter lifespan of the cell [4]. The capacity degradation rate of Li-ion batteries also increases significantly when operate at low temperatures. At −10 °C, the capacity loss of a Li-ion cell can reach 25% after only 40 cycles when the charge rate is 0.5*C* [5] Thus, keeping the lithium-ion cells at a certain ideal temperature range is important to improve the performance of the battery pack.

Many heating techniques have been proposed to pre-heat the battery packs in frigid climates, and can be categorized into two main groups: external heating and internal heating. External heating methods such as air heating [6], liquid heating [7], heat pipe heating [8], and burner heating [9], etc., are known to provide high reliability at low complexity [10]. However, their main disadvantages are the low temperature uniformity among the cells, low heating efficiency due to heat transfer loss, and significant weight increases of the system [10]. On the other hand, internal heating methods utilize the high cell impedance at low temperatures to generate heat from inside the cells using the charge or discharge current that directly warms up the cell instead of passing through the cell surface to the interior. One of the internal heating methods is internal self-heating where the cell is discharged using a constant current or constant voltage to warm up the cell [11]. Alternating current heating was also proposed where an

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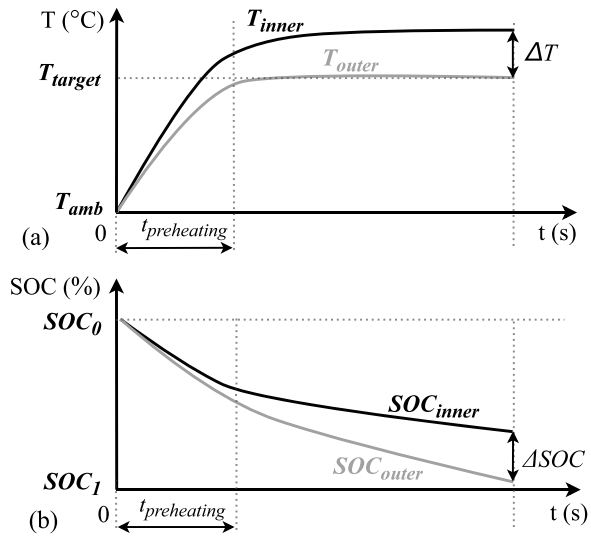

Fig. 1. The variances in (a) temperature or (b) SOC distribution between two layers during different heating processes when only considering SOC balancing or temperature balancing, respectively.

alternating current is used to provide a high current to the cells while minimizing the energy loss from just discharging the cells [12–14]. Another internal heating method is mutual pulse heating where a group of cells is used as a temporary reservoir for exchanging electrical energy during the charging/discharging of the other group [15]. In general, internal heating methods offer shorter heating time while consuming less energy, and incurring less weight [10].

Despite having many advantages compared to external heating methods, internal heating methods often rely on the precise electrical characterization of the cells to determine the optimal heating strategy as in [12–14]. In practice, even with the same chemistry and physical size, lithium-ion batteries can still have significant variance in nominal capacities, self-discharge rates, and especially differences in the internal impedance [16,17]. Achieving balanced SOC and uniform temperatures has been shown to be critical to ensure high performance of the pack, but remains a challenge when using internal heating methods [18]. Additionally, previous heating methods assume that the battery pack will be used immediately and do not consider maintaining the temperature of the battery pack after pre-heating. As the heating duration often varies from several minutes up to half an hour [19], the immediately usage after pre-heating is not always guaranteed. If it is not actively heated, the cell temperatures will start dropping and the heating process will be needed again before the cells reached the ideal operating temperature. In this study, we propose a control mechanism using mutual pulse heating that ensures the uniform temperature distribution and the balanced SOC among the cells in the pack throughout the heating process.

The remaining of this study is organized as follows. We discuss the internal heating method that is utilized in this study along with the effects of temperature distribution and unbalanced SOC on the performance of battery packs in Section 2. The construction of electrical and thermal models for battery cells is detailed in Section 3. The proposed heating technique and a practical battery pack structure are presented in . The evaluation of the heating technique are discussed in Section 5. Finally, we conclude our work in Section 6.

2. Overview

2.1. Problems of simultaneous temperature and SOC balancing

Lithium-Ion battery packs are usually constructed from multiple cells where they are arranged in various pack structures with multiple layers of cells that expose to the environment [10,20,21]. Because of having different amounts of surface area exposed to the environment where the rates of heat loss are not the same [22], they have been shown to have poor temperature uniformity during discharging as the cells heat up internally and the temperature variance among the layers can be as large as 35 °C for a rectangular battery pack that made of 18650 cells [20]. Similarly, during the heating process of the batter pack, the temperature rise among the inner cells is larger than that of the outer cells, resulting in nonuniform temperature distribution as shown in Fig. 1(a). This difference in temperature distribution can accelerate the degradation process that reduces the nominal lifespan of lithium-ion battery cells as the cell with higher temperature among multiple cells connected in a parallel topology is likely to experience a larger discharge current during operation [23,24]. On the other hand, when heating the battery pack where the temperature uniformity is the primary goal, the outer layer will be heated more often than the inner layer, resulting in a significant imbalance in SOC after pre-heating as depicted in Fig. 1(b). The differences in SOC can cause inconsistency in the performance of the cells which accelerates the aging process [25]. It was shown that a variation in temperature of 8 °C can cause a current deviation of 14%, while even only 8% of SOC variation can result in a current difference of 62% among the cells [26].

Previous studies on battery heating methodologies usually focus on heating a single cell or a small group of cells. However, when an optimal heating method is used without careful considerations on thermal characteristics of the battery pack, the performance of the pack would be severely degraded.

2.2. Balancing temperature and state-of-charge of battery packs

As thermal non-uniformity and SOC imbalance negatively affect the battery cells in many aspects, many works have been done to address each of the problems separately or both. An passive SOC balancer was proposed to equalize the SOC among cells by over-charging the cells or burning the excess charge through shunt-resistors [27]. This method is relatively simple to implement, but has a low efficiency. Another study proposed an active SOC balancer where external switching circuits are used to transfer the energy between the cells and external storage elements such as capacitors and inductors [28]. This method is more effective compared to the passive SOC balancer but requires more advanced control algorithms. Meanwhile, a common technique for balancing the temperature among the cells is active cooling with unidirectional coolant flow where air or liquid coolant can be employed [22]. This technique is simple to implement but it cannot resolve the imbalance temperatures originated from the variations in cell resistance. The nonuniform temperature can also be reduced by balancing losses among battery cells using load scheduling where individual switching circuits are used at cell or power unit level to discharge the cells according to the cell temperatures [25]. Recently, a simultaneous thermal and SOC balancing technique was proposed where each power unit is equipped with a multilevel converter for optimally shifting the power losses among the cells depending on their electrical and thermal properties [29]. This technique provides initial insight on thermal and SOC balancing, but requires the main load to be used in order to close the energy flow while also having high computational complexity. These studies have demonstrated that temperature distribution, or SOC, or both can be balanced at the same time. However, most of them solely focus on solving these problems during runtime when the cells are being used to avoid thermal-runaway that is fundamentally different compared to heating the cell from subzero temperatures since the electrical properties of the cell are not the same. They also often require precise characterization of the cells, along with complex computation to keep track of system states during the balancing process which pose a significant overhead.

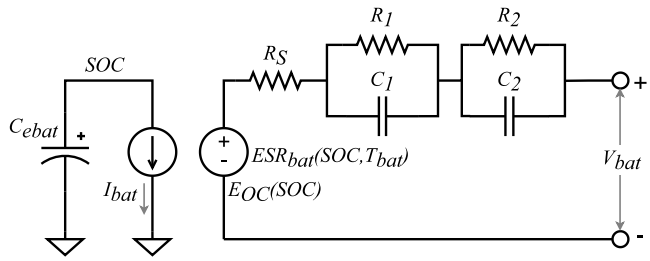

Fig. 2. Equivalent circuit model for a Li-ion battery.

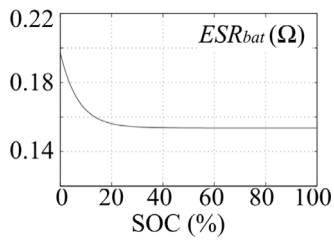

Fig. 3. Equivalent internal resistance of a Lithium-ion cell at different SOC levels [12].

2.3. Simultaneous internal heating using mutual pulse heating

As mentioned in Sections 2.1 and 2.2, the problems of simultaneous temperature and SOC balancing during heating in subzero temperatures remain unsolved. The main challenges lie in the lack of a reliable control policy and the complexity of system modeling since the electrical properties of the cells change over time due to the aging process. As an active SOC balancer was used in [28] to equalize the SOC of the cells with high efficiency and short heating time while incurring little overhead, our observation is that mutual pulse heating [15], which is also similar to the active SOC balancer, can be utilized to achieve temperature uniformity by strategically activating the heating process for each layer of cells independently. Different from [28], mutual pulse heating is a pulsed heating method that employs different groups of battery cells as the energy reservoirs during the heating process that increases the utilization of the transferring current to heat up the cells faster while minimizing the losses. While it provides a fast heating time, the effects of pulsed heating on the battery capacity was examined in details in [30]. It was shown that the pulse heating method does not significantly impact the life span of the cells in long-term when the pulse amplitude and duration are chosen properly. Taking the advantages of mutual pulse heating, this study presents a heating technique that simultaneously balances the SOC and ensures the uniform temperature distribution during and after the pre-heating while requiring only the knowledge of the temperature and SOC level of the cells. The proposed technique is discussed in details in Section 4.

3. Battery models

This section presents the electrical and thermal dynamic models of a lithium-ion battery cell. These models are needed to provide an accurate evaluation for the performance of the proposed heating technique.

3.1. Electrical model

In internal heating techniques, the heat is generated inside the battery cell. It is necessary to understand the behaviors of the cell when being heated internally in sub-zero temperatures. An equivalent electrical model of a Li-ion cell was comprehensively analyzed in [31] and is shown in Fig. 2.

In this model, a full-charged capacitor, C_{ebat}, represents the nominal electrical of the cell. This capacitor charge level corresponds to the SOC of the cell and can be charged/discharged by applying a current on the terminal voltage, V_{bat}. The passive components, R_S, R_1, and C_1 form the internal impedance of the cell. When the cell provides electrical power to a load, there is energy loss depending on the load current, I_{bat}. The energy loss is translated to the heat on the passive components and contribute to the temperature rise of the cell. This process is elaborated further in Section 3.2.

The SOC of the cell can be estimated by integrating I_{bat} over time [32]:

$$\mathrm{SOC} = \mathrm{SOC}(0) - \frac{\int_0^t I_{bat}(t)dt}{Q_{bat}}, \tag{1}$$

where $SOC(0)$ is the initial SOC in percentage and Q_{bat} is the nominal battery capacity in Ah.

The open-circuit voltage of a typical Li-ion cell, E_{OC}, was investigated in [33], and is given by the following nonlinear equation:

$$E_{OC} = b_{11}e^{b_{12}\mathrm{SOC}} + b_{13}\mathrm{SOC}^4 + b_{14}\mathrm{SOC}^3 + b_{15}\mathrm{SOC}^2 + b_{16}\mathrm{SOC} + b_{17}, \tag{2}$$

where b_{ij} are regression coefficients and were extracted from [34] according to the extraction method in [33].

The internal impedance of the battery cell was examined for multiple lithium-ion cells in [35,36], and showed that the internal impedance decreased less than 10% across SOC from 0% to 100%. Moreover, from around 25% of SOC and above, the cell impedance remains relatively consistent regardless of SOC changes for different types of battery cells including the beginning-of-life and aged cells [36]. The internal impedance as a function of SOC was examined in [37] and is shown in Fig. 3. Meanwhile, battery heating is often used when the SOC is full while it only consumes a small percentage of energy. Therefore, the SOC can remain relatively high during the heating process. We assume that the SOC is always above 25% during the process, and ignore the dependency of the cell internal impedance on the SOC in this study.

The cell impedance was inspected in recent studies. In [38], resistors R_1 and R_2 in Fig. 2 remain unchanged during discharging in a wide range of temperatures from −20 °C to 5 °C while the serial resistance, R_s, becomes the dominant part of the cell impedance. The heat generation was calculated in [39,40] using a single lumped internal resistance, R_{bat}, that includes the heat dissipation through R_s and R_1-C_1. The internal impedance was also evaluated in [41] for charging/discharging at different SOC indicating a similar impedance–temperature dependency in a wide range of temperatures. Accordingly, we evaluated the single lumped internal resistance of a 2500-mAh Samsung SDI 18650 25R (T11-H2500S) cell in a wide range of temperature from −20 °C to 90 °C by discharging the cell inside a temperature chamber. The measurement of internal resistance of the cell using a temperature chamber (Coretech HQ-DTH-81) for varying the temperature and a programmable electronic load (Maynuo M9716) for changing the discharge current is illustrated in Fig. 4. We observed that the cell internal resistance remains consistent when the cell temperature is larger than 10 °C. However, at lower temperatures, the internal resistance is not only strongly dependent on the cell temperature, but also varies greatly when the cell is discharged at different rates as showed in Fig. 5.

A multi-variable three-degree polynomial regression model is used to describe the dependency of the cell internal resistance on the discharge current and the cell temperature, and is expressed as:

$$\begin{aligned} R_{bat} = {} & p_{00} + p_{10}T_{bat} + p_{01}I_{bat} + p_{20}T_{bat}^2 + p_{11}T_{bat}I_{bat} \\ & + p_{02}I_{bat}^2 + p_{30}T_{bat}^3 + p_{21}T_{bat}^2I_{bat} + p_{12}T_{bat}I_{bat}^2, \end{aligned} \tag{3}$$

where T_{bat} is the cell temperature in degree Celsius, I_{bat} is the discharge current in Ampere, and p_{ij} are regression coefficients, and are listed in Table 1.

The complex dependency of the cell internal resistance on temperature and discharge current indicates that exclusively relying on

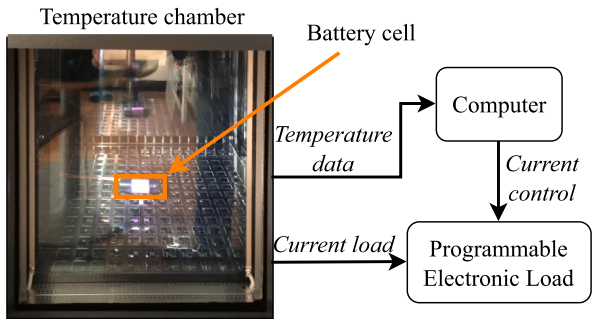

Fig. 4. Internal resistance measurement for a Li-ion cell in a temperature controlled chamber.

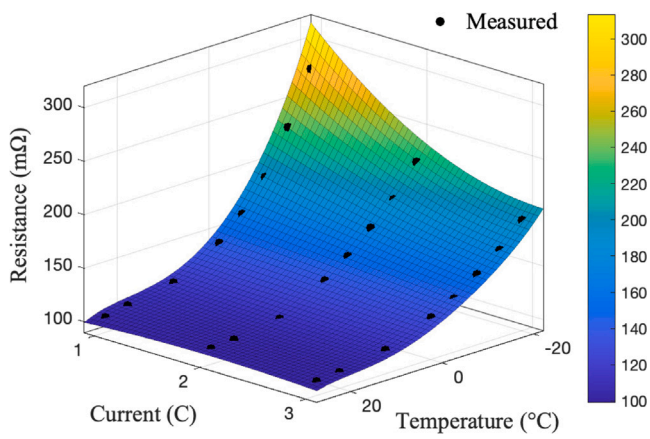

Fig. 5. Internal resistance of a Lithium-ion cell at different discharge rates in sub-zero temperatures.

Table 1
Electrical parameters for battery models.

Parameters	Values	Parameters	Values
Q_{bat}	2.5 Ah	b_{16}	1.6544
b_{11}	−0.2653	b_{17}	3.3564
b_{12}	−61.6492	b_{21}	0.0435
b_{13}	−2.0398	b_{22}	−14.2753
b_{14}	5.2765	b_{23}	0.1537
b_{15}	−4.1733	p_{12}	−0.0429
p_{00}	158.6	p_{10}	−5.633
p_{01}	−8.659	p_{20}	0.1532
p_{11}	0.8242	p_{02}	0.6468
p_{30}	-8.114×10^{-4}	p_{21}	−0.0141

a specific electrical profile of the cell for the heating process is not reliable and can create unexpected results. In this study, this dependency is only used in simulations for demonstrating the feasibility of the proposed heating technique. The operation of the heating technique without using this electrical profile is discussed in Section 4.

3.2. Thermal model

To evaluate how heat generation from one battery cell can affect the temperature changes of that cell and surrounding cells, the thermal models for a single battery cell and the battery pack are needed.

A typical battery pack consists of cylindrical 18650 cells with the casing made of thermoplastic polymer is presented in Fig. 6. The battery pack is divided into two layers based on the rates of heat loss from the individual cells to the outside environment. In this pack model, we assume that the space between the cells is filled with air. The pack cover provides protection and acts as a thermal insulation layer that reduces the heat loss from convective heat transfer.

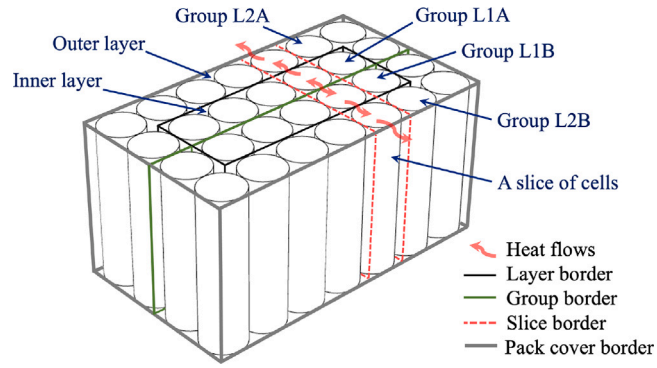

Fig. 6. A typical pack geometry using Li-ion 18650 cells with heat flows and group division.

The temperature distribution inside 18650 cylindrical cells during self-heating was investigated in [42]. The difference between the inner core temperature and the outer temperature of the cell is small at about 0.6 °C at 2C, and is moderate around 2 °C at 4C discharge rate. Since the proposed maximum discharge rate of the cells in this study is 3C as discussed in Section 5, this difference is rather trivial in our model. Moreover, when the temperature rise and the rate of heat loss are slow during the heating process, the Biot number which represents the ratio of the thermal resistances inside and at the surface of the 18650 cell is expected to be reasonably small that the cell can be considered as a lumped system of uniform temperature [43]. The significant gradient inside the cell only exists when the rate of thermal changes is high which is not the case in our models. Thus, we use a single parameter, T_{bat}, to represent the cell temperature.

The increase of the T_{bat} is dependent on the internal heat generation of the cell, P_{gen}, and losses to various factors. This relationship was examined in [44] and is expressed as:

$$mC_{bat}\frac{\partial T_{bat}}{\partial t} = P_{gen} - h_b A_{bs}(T_{bat} - T_{amb}) - \varepsilon\sigma A_{bs}(T_{bat}^4 - T_{amb}^4) - Q_{ext}, \quad (4)$$

where C_{bat} is the heat capacity of a battery cell, m is the cell mass, h_b is the heat transfer coefficient between the battery and surrounding air, A_{bs} is the battery surface area, σ is the Stefan–Boltzmann constant, ε is the battery surface emissivity, T_{amb} is the ambient temperature, and Q_{ext} is external heat conduction from the cell to the connectors. The right-side elements in Eq. (4) represent the heat gain, the losses through convective, radiation, and conductive heat transfer, respectively. In frigid climates, the loss through radiation heat transfer is trivial compared to convective heat transfer [45], and hence, it is ignored in this study where the ambient temperatures are low.

The battery pack was divided into layers where the differences in the cell temperature in the same layer are negligible, and thus, we ignore the conduction heat transfer from one cell to another through the connectors.

The total amount of heat change in a single lithium-ion cell, P_{gen}, consists of the heat from overpotential, Q_p, and the heat from entropy change, Q_s, and is expressed as follows [46]:

$$P_{gen} = Q_p + Q_s, \quad (5)$$

$$Q_p = I_{bat}^2 R_{bat}, \quad (6)$$

$$Q_s = I_{bat} T_{bat} \frac{\partial E_{oc}}{\partial T_{bat}}, \quad (7)$$

where the heat Q_p is exothermic in both charging and discharging while the heat Q_s is exothermic during discharging and endothermic during charging [47,48]. Meanwhile, Q_p typically exceeds Q_s in sub-zero temperatures [45]. Thus, the total generated heat is positive during both processes, and the cell temperature increases in both groups in mutual pulse heating.

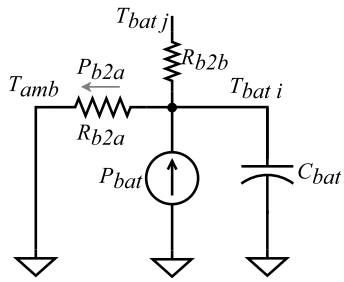

Fig. 7. Lumped RC-thermal circuit model for a unit cell.

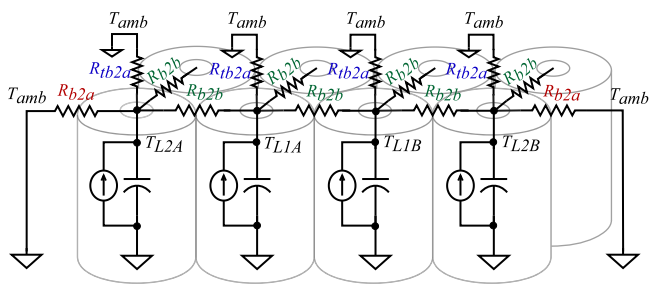

Fig. 8. Thermal RC network model for a slice in the battery pack.

The heat transfer contributing to the temperature change in a single cell and various loss paths are illustrated in Fig. 7. The cell temperature increases when the heat capacitor C_{bat} is charged from P_{gen}. Similarly, the cell temperature decreases when C_{bat} begins to lose heat through thermal resistors.

We assume that the thickness distribution of the thermoplastic polymer casing is uniform to ensure the same thermal dissipation rate for the whole battery pack. The thermal resistance of the cover layer that represents the heat loss from a battery cell to the environment through the cover layer is defined as:

$$R_{b2a} = \frac{h_{cover}}{\lambda_{Poly} A_s}, \tag{8}$$

where λ_{Poly} is the polycarbonate thermal conductivity, h_{cover} and A_s are the thickness and the corresponding area of the cover layer, respectively [49]. The thermal conductivity for polycarbonate was examined in [50] and is presented in Table 2. The heat loss from one cell to another lower-temperature cell through the air inside the pack is illustrated by the thermal resistance, R_{b2b}, and was acquired from [46].

The heat transfer in the battery pack is analyzed using a slice of battery that consists of cells from four groups of two layers and is shown in Fig. 8. In this model, the heat dissipations R_{b2a} is used to calculate the heat loss from the outer cells to the environment while R_{tb2a} is used to evaluate the loss from the top and bottom area of all the cells to the environment.

From (4) and (8), the dynamic thermal model of one cell for multiple-layer heat transfer that describes the changes in temperature of a cell depending on the temperature of neighboring cells and ambient temperature can be written as:

$$P_{gen_i} = mC_{bat}\frac{\partial T_{bat_i}}{\partial t} + \frac{T_{bat_i} - T_{amb}}{R_{b2a}} + \frac{T_{bat_i} - T_{amb}}{R_{tb2a}} + \sum_{j=1}^{N}\frac{T_{bat_i} - T_{bat_j}}{R_{b2b}} \tag{9}$$

where bat_i represents the primary cell that is being evaluated, and bat_j represents the surrounding cell that is affected when there is a difference in temperature between two cells. The right-side elements represent the change in cell temperature, the losses to the ambient through the outer sides, and the change in temperature affected by the nearby cells, respectively. Similarly, the temperature change of any given cell in a slice can be obtained.

Table 2
Mechanical and thermal parameters for battery models.

Mechanical model			
Parameters	Values	Parameters	Values
m_{bat}	0.048 kg	h_{bat}	65×10^{-3} m
d_{bat}	18×10^{-3} m	h_{cover}	2×10^{-3} m
A_s	1.17×10^{-3} m^2	A_{tb}	6.48×10^{-4} m^2
Thermal model			
Parameters	Values	Parameters	Values
λ_{Poly}	0.22 W/(mK)	C_{bat}	1109 J/(kg K)
R_{b2a}	19.39 K/W	R_{b2b}	10.84 K/W
R_{tb2a}	34.402 K/W		

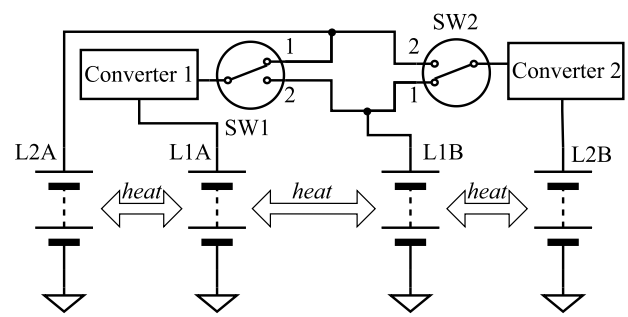

Fig. 9. Electrical connection between groups of cells in a battery pack.

4. Simultaneous temperature and SOC balancing

4.1. Battery pack organization

As discussed in Section 3.2, the pack is divided into two layers, namely the inner layer and the outer layer. Each layer is further divided into two subgroups with the same thermal and electrical characteristics, resulting in two subgroups of the inner layer, L1A and L1B, along with two groups of the outer layer, L2A and L2B. This separation is needed because the groups will be used as the temporary energy storage to achieve both temperature and SOC balance during the heating process. We assume each subgroup consists of one cell, but in practice, a subgroup can be a power cell (PC) that contains multiple cell in different configurations depending on the application.

These four groups of cells are connected strategically using two switches and two DC–DC power converters as shown in Fig. 9. The switches are used to connect the groups within a layer or interconnect groups between different layers depending on the operating state of the proposed technique. In this model, two cascaded bidirectional boost converters are used. The complexity of these converters has been shown as trivial, and the conversion efficiency can be as high as 99% [51]. In this study, we set this efficiency at only 90% to resemble the operation of the proposed technique in real-world conditions.

In practice, the cells in each group can be connected in serial to reduce the complexity of electric connections in the pack. However, balancing circuits are still needed to ensure uniform voltage distribution for every cell in the pack. In this study, we analyze the model in one slice with one cell for each group to simplify the heat transfer process. The behaviors in terms of heat distribution and electrical transfer are similar for almost all of the slices in the pack except for two outer slices which would have slightly different thermal properties due to having a larger surface area exposed to the air.

4.2. The proposed heating technique

To simplify the representation of various temperature variables in the model, T_{inner} and T_{outer} are used to denote the average temperature between groups of the inner layer and the outer layer, respectively.

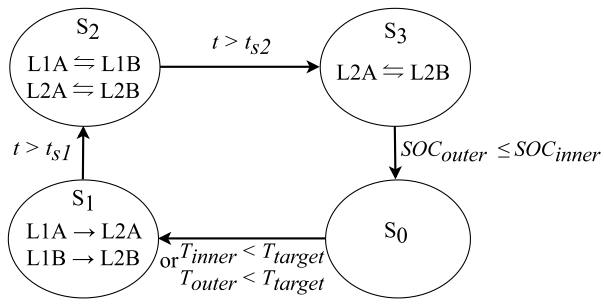

Fig. 10. Energy transfer in four states of the proposed heating technique.

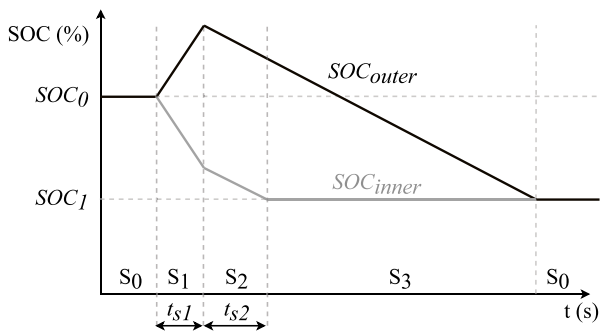

Fig. 11. SOC balancing of two layers in an incremental heating cycle.

Similarly, SOC_{inner} and SOC_{outer} denote the average SOC between groups of the inner layer and the outer layer, respectively.

During the heating process, due to having a larger area exposed to the outside environment, the outer layer will lose more heat than the inner layer. Thus, the heating duration for the outer layer is expected to be larger than that of the inner layer. However, a larger heating duration of the outer layer will result in a higher SOC reduction. To tackle this problem, we use mutual pulse heating in a four-state process where the energy is transferred between the groups as shown in Fig. 10. In this process, the outer layer will be distributed with a higher average SOC in short duration to extend the heating time while minimizing the temperature differences between two layers at the end of each heating cycle. The heating process for the battery pack is performed in the following states:

- S_0: The idle state. The system is off and there is no heating activity.
- S_1: Two switches SW1 and SW2 are at their first position to connect the groups of the inner layer to the groups of the outer layer. The energy is transferred from the inner layer to the outer layer for all of the groups using one-way mutual pulse heating.
- S_2: Two switches SW1 and SW2 are at their second position to connect the groups in the same layer. A bi-directional mutual pulse heating process is performed between two groups of the same layer.
- S_3: The switches SW1 and SW2 are still at their second position. A bi-directional mutual pulse heating process is used for the outer layer only. The heating process is maintained until the average SOC are balanced.

The changes in SOC during the process are illustrated in Fig. 11. In the energy transfer state, S_1, pulse currents are used to avoid the long-charging duration. We observed that a few seconds are enough to make a significant difference in SOC in this state. The heating state, S_2, is used to increase the temperature of both layers. Because SOC_{outer} is now higher than SOC_{inner}, the outer layer can be safely heated for a longer duration until the two average SOC are at a similar level. This solves the problem of imbalance SOC at the end of each heating cycle. To achieve the temperature balance, we use Algorithm 1 to adjust the duration of t_{s1} and t_{s2} for S_1 and S_2, respectively.

The algorithm uses the temperature values of T_{inner}, T_{outer}, and the target temperature, T_{target}, as the input. It starts by assigning initial values for t_{s1} and a time coefficient, k. After each heating cycle, if both layer temperatures are smaller than T_{target}, t_{s1} and k will be reduced by one unit. Otherwise, if both layer temperatures are larger than T_{target}, t_{s1} and k will be increased by one unit. Finally, t_{s2} is determined as the product of k and t_{s1}. The new values of t_{s1} and t_{s2} will be used in the next heating cycle, and the temperature of both layers can be balanced.

Algorithm 1 Time scale tuning

Input: T_{inner}, T_{outer}, T_{target}
Output: t_{s1}, t_{s2}

```
procedure TUNING
    Set initial values t_s1 = 1, k = 10
    if T_inner < T_target and T_outer < T_target then
        t_s1 = t_s1 + 1 and k = k + 1
    if T_inner > T_target and T_outer > T_target then
        t_s1 = t_s1 − 1 and k = k − 1
    Set t_s2 = k × t_s1
    return t_s1, t_s2
```

It is important to note that this algorithm does not use the equivalent internal resistance of the cell as the input variable. Instead, only temperature and SOC of the cell are needed. In practice, the cell temperatures can be measured directly using external thermal probes, and SOC can be estimated from the terminal voltage of the cells. Thus, it also works for other types of lithium-ion batteries with different electrical profiles.

5. Evaluation

5.1. Temperature and SOC imbalance in mutual pulse heating

In this section, we use the previously constructed electrical and thermal models to verify the limitations of using mutual pulse heating without considering the thermal distribution in a battery pack with multiple cells. The mechanical, electrical and thermal parameters for a standard Lithium-Ion 2.5Ah $LiNiMnCoO_2$ 18650 were examined in [12, 52] and are listed in Tables 1 and 2.

Two bi-directional boost converters are employed in this model. In a conventional boost conversion, the input side with lower voltage always has a higher-rate current compared to the output side with higher voltage and lower-rate current as depicted in Fig. 12(b). As a result, the amount of heat generated internally at the discharge group is higher than that of the charge group. If the duration of the charge/discharge interval is not consistent, this effect can lead to uneven temperature gains among the cells. This also results in a higher-rate current at the terminal of the cell in the discharge group and lower voltage gain at the terminal of the cell in the charge group as shown in Fig. 12(a).

In the heating process, we limit the discharge current of the cell such that the terminal voltage is never below 2.8 V during the heating process to minimize the effect of lithium plating. As the cell temperature rises, its internal resistance gets smaller and the voltage drop at the terminal becomes less significant. For maintaining the same or higher heat generation inside the cell, a higher charge/discharge current is needed. It was shown that pulsed heating using 4C current with the pulse interval of 1.6 s in 800 h only reduces the cell capacity by 1% which is trivial compared to the heating duration [30]. Thus, the current rate of 3C and pulse interval of 1 s are used in our evaluation models to minimize the degradation of cell capacity from the self-heating process.

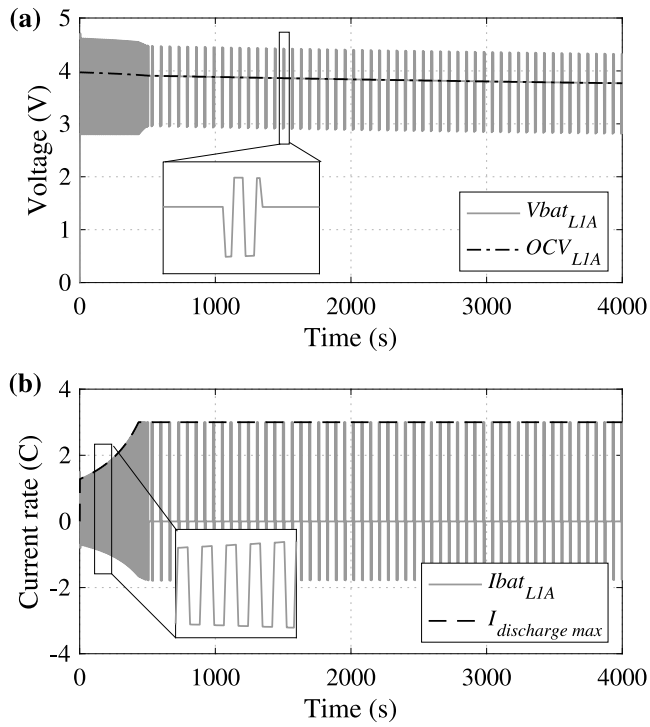

Fig. 12. (a) Voltage and (b) current responses from one cell of the group L1A in mutual pulse heating.

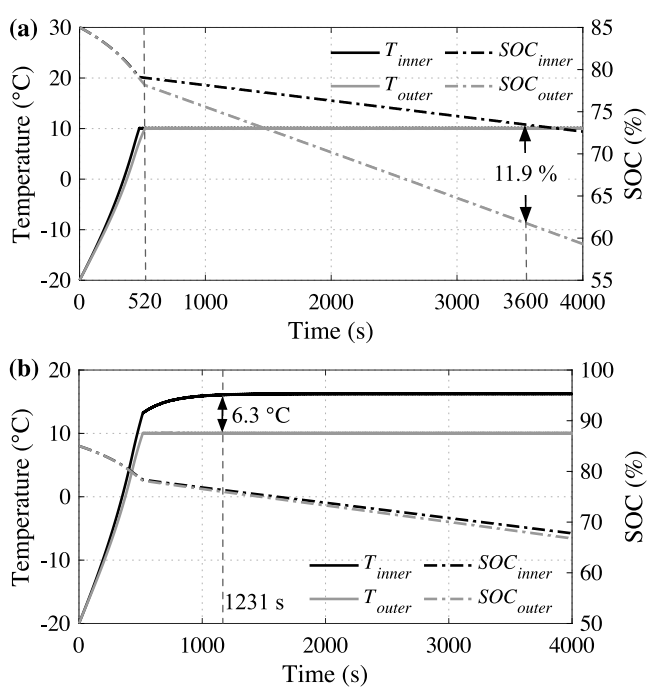

Fig. 13. Heating process for the battery pack when only considering (a) temperature balancing or (b) SOC balancing.

The heating results from −20 °C to 10 °C for a slice of cells using mutual pulse heating when considering only temperature balancing are presented in Fig. 13(a). During heating, because the outer layer has a larger exposed area to the environment that creates a higher rate of heat loss, it is heated in a longer duration, resulting in a small difference

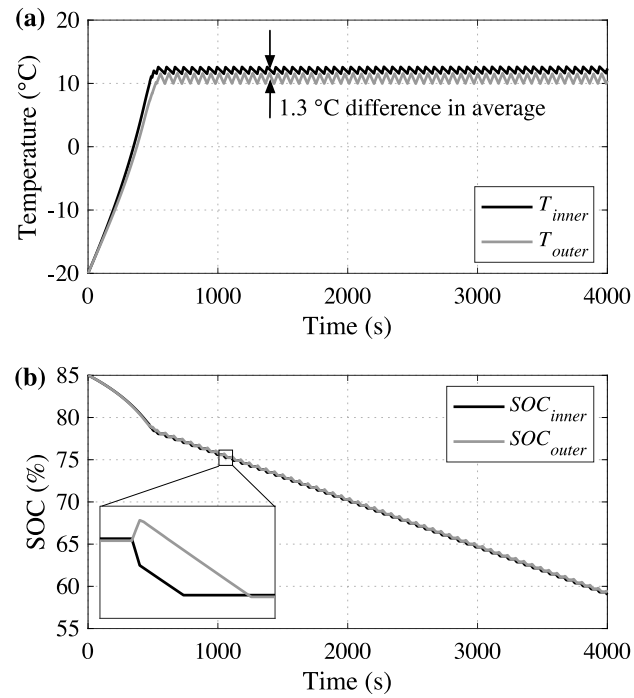

Fig. 14. (a) Temperature and (b) SOC responses during and after pre-heating using the proposed technique when the target heating temperature is 10 °C.

in SOC of 0.9% at t = 520 s. However, this difference is exaggerated up to 11.9% after one hour of heating. Meanwhile, when performing mutual pulse heating with only SOC balancing, the temperature difference between two layers is shown in Fig. 13(b). The temperature difference is only 3.2 °C right after the pre-heating at t = 518 s, and is increased to 6.3 °C after 1231 s. This temperature difference is larger than 5 °C which is the recommended threshold for the safe operation condition of the cells [53] In general, these differences are obvious upon the completion of the pre-heating process. Meanwhile, they can be enlarged significantly as the heating time prolongs. If the SOC or temperatures among the cells are imbalanced at the beginning of the process, the differences can be exaggerated further [12].

5.2. Heating process using the proposed heating technique

In this section, the proposed heating technique is used for the heating process with the same condition. The results of temperature and SOC responses using the proposed heating technique are presented in Fig. 14. During the heating process, the temperature rise in the pre-heating process was similar to that of the previous scenarios. However, in Fig. 14(a), when the pre-heating is completed at t = 521 s, there is only a small difference of 1.3 °C in the average temperature distribution between the two layers after the pre-heating, and it is maintained consistently over time. Meanwhile, the SOC difference is only 0.4% during the whole process as showed in Fig. 14(b). The SOC pattern in each heating cycle after the pre-heating process is similar to that of the four-state process as shown Fig. 11. Throughout the heating process, both temperature distribution and SOC are balanced consistently. As the control algorithm only uses the temperature and SOC of the cells from the groups to manage the process regardless of the ambient temperature, these results show that the proposed technique is resilient to variations in cell resistance and the rate of heat loss while only adding a minimal complexity. This also means the proposed technique can be used for battery packs that made from other types of battery cell rather than cylindrical cells as long as the variations of temperature among the cells are significant.

Table 3
Effects of ambient temperature on the heating performance when heating to 10 °C.

T_{amb} (°C)	Temperature difference between T_{inner} and T_{outer} (°C)	SOC reduction during preheating	Pre-heating duration (s)
−30	1.8	9.2%	810
−25	1.5	7.5%	611
−20	1.3	6.1%	463
−15	1.0	4.9%	348
−10	0.7	3.7%	257
−5	0.5	2.8%	182
0	0.3	1.8%	115

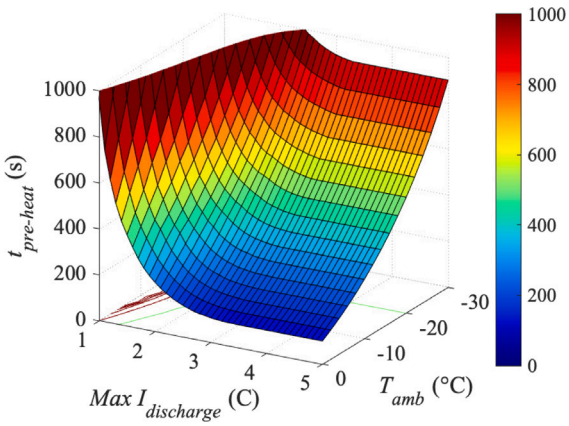

Fig. 15. Dependency of the pre-heating time on the maximum discharge current and ambient temperature when the target heating temperature is 10 °C.

5.3. Effects of ambient temperatures on the heating performance

The results from the proposed heating technique showed significant improvements in the heating process. However, a small difference in temperature distribution between the two layers still exists. As shown in Table 3, the average difference in temperature between two layers is 1.8 °C when the ambient temperature is −30 °C, and tends to get smaller as the ambient temperature increases. This result showed the robustness of the technique in keeping the temperatures stable during the heating process across a wide range of ambient temperatures.

The pre-heating duration and SOC usage during the pre-heating process using the proposed heating technique were also examined. In Table 3, both the SOC usage and the pre-heating time decreases significantly as the ambient temperature increases.

In the battery heating process, shortening the pre-heating time is always desirable. One simple method is increasing the heating current. However, as shown in Fig. 15, increasing the maximum discharge current above 3*C* does not help in reducing the pre-heating time in any given ambient temperature. Thus, choosing the optimal current for mutual pulse heating is necessary to achieve a faster heating time while also minimizing negative effects on the battery lifespan.

6. Conclusions

In battery-powered applications, heating the battery in frigid climates is a critical task. However, heating a medium to large-size battery pack sill remains a challenge because of the non-uniform temperature distribution and the imbalance SOC among the cells due to the variations in electrical and thermal characteristics of the battery cells. In this study, a heating technique was proposed to accomplish both objectives. The battery pack was divided into four groups to create temporary reservoirs for storing energy during a four-state process in each heating cycle. The evaluation results with realistic electro-thermal parameters showed that the proposed heating technique can keep the SOC relatively identical while the difference in temperature among the groups of cells was below 1.8 °C throughout the heating process in a wide range of sub-zero temperatures. The proposed technique also does not require the precise characterization of the cell internal impedance as the heating duration was adjusted adaptively at any ambient temperature. By rerouting the electrical flow in a simple way, this heating technique can be integrated into a thermal management system to improve the performance and extend the lifespan of the battery pack.

CRediT authorship contribution statement

Hien Vu: Conceptualization, Methodology, Data curation, Visualization, Investigation, Writing – original draft. **Donghwa Shin:** Supervision, Reviewing and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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